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Unveiling Underwater Structures: MobileNet vs. EfficientNet in Sonar Image Detection

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Abstract

Deep learning techniques have led to an increased use of Convolutional Neural Networks (CNN) in recognizing images for marine surveys and classifying underwater objects. Applying CNN for the automatic identification of targets in side-scan sonar (SSS) images can boost accuracy and efficiency. Using transfer learning and deep learning as lenses, this work explores the field of underwater item detection. Comparing performance of MobileNet and EfficientNet in performing the task of underwater item classification from sonar images is the central research objective of this study. These two well-known convolutional neural networks were the subject of a thorough investigation to help explain this.In order to fine-tune pre-trained models on a dataset of sonar images, the study approach used the transfer learning technique. We conducted tests on MobileNet and EfficientNet and thoroughly assessed each of their accuracy levels.This study has two main conclusions. First, showing promise as a strong model for underwater object identification, EfficientNet performed exceptionally well in the classification challenge.

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Keywords:
Side-Scan Sonar; Convolutional Neural Networks; Transfer Learning; MobileNetV3; EfficientNetV2;

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1. Introduction

Underwater object classification technology is a vital blend of various fields such as sonar technology, signal detection theory, artificial intelligence, pattern recognition, and computer technology. As sonar imaging technology advances, the categorization of underwater objects from sonar images becomes a significant area in ocean development. Sonar images contain object-highlight regions, shadow regions, and sea-bottom reverberation regions. These regions are crucial for understanding what is in the underwater environment, but sonar images often have low contrast, blurry edges, and high noise, which poses a challenge for accurate classification. In recent years, the field of geology has seen the integration of artificial intelligence (AI) algorithms in diverse applications. These applications encompass seafloor sediment accumulation prediction, oil and gas reserve forecasting, seafloor substrate classification, geological drilling, and geological hazard prediction. Notably, Convolutional Neural Networks (CNN) have been extensively utilized for the recognition of targets within side-scan sonar (SSS) images.Leveraging CNN for the automated recognition and prediction of SSS images has the potential to significantly enhance both efficiency and accuracy in these geological applications. The scarcity of labeled datasets is a significant obstacle to underwater item classification for geological applications. Data is scarce since obtaining and annotating SSS images is expensive and complicated. Moreover, the dynamics and complexity of the undersea environment provide challenges to the development of reliable and broadly applicable categorization models.Underwater object categorization technology has the potential to revolutionize geological applications in spite of these obstacles. CNN-based classification models, for instance, may be used to: Automate the recognition of geological characteristics including undersea dangers, silt kinds, and seabed faults. Boost the precision and effectiveness of geological mapping and surveys. Encourage the search for and exploitation of maritime resources.Improved monitoring and management of coastal habitats. In order to tackle the issues provided by limited data and the requirement for effective and deployable models, the proposed research assessed the performance of two lightweight CNN models for classifying underwater objects: MobileNet and EfficientNet. A family of CNN models called MobileNet is intended to be lightweight and efficient, making it appropriate for use on mobile smartphones and other edge devices. A series of CNN models known as EfficientNet was created with scalability and efficiency in mind. This allows the models to accomplish cutting-edge accuracy on a variety of tasks with little computing complexity.

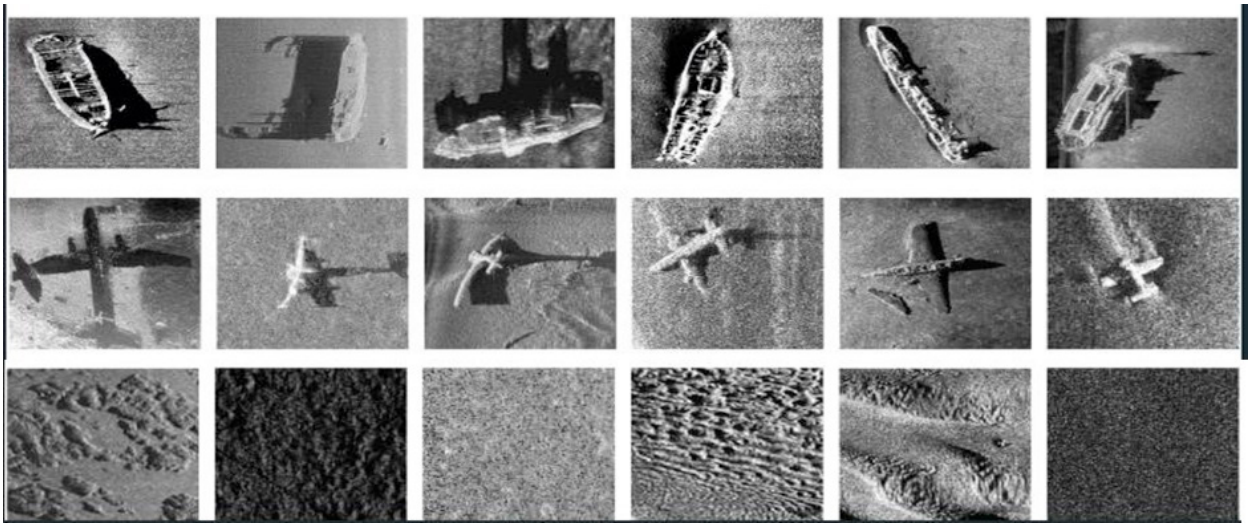


Fig. 1: SeabedObjects-KLSG II

2. Literature Review

Xing Du, Yongfu Sun, et al. [1] shows convolutional neural networks (CNNs) have recently been demonstrated to perform very well for underwater object categorization tasks. MobileNet and EfficientNet are two of the most wellliked

CNN designs for mobile and embedded devices. The performance of four different CNN models, namely AlexNet, VGG-16, GoogleNet, and ResNet101, for underwater object classification in side-scan sonar images is compared. The impact of transfer learning on the effectiveness of these models was also assessed by the authors. The findings revealed that AlexNet had the lowest accuracy while GoogleNet had the best accuracy. All four models showed an increase in accuracy thanks to transfer learning, but ResNet101 showed the most gain.

Keshetti Sreekala, N. Nijil Raj, et al. [2] uses deep convolutional neural network (DCNN) technique for underwater image preprocessing and object recognition is presented. The DCNN is trained to acquire the illumination plot and to merge both max-RGB and shade-of-grey techniques to increase underwater vision. It is demonstrated that this technique works well for overcoming the issue of underwater images with poor lighting. A Kalman filter (KF) technique for tracking and detecting submerged objects is also presented in the research. The tracking and detection system's precision and resilience are increased by fusing the DCNN data with the inertial measurement unit (IMU) info from the AUV using the KF. Simulation results demonstrate that the suggested DCNN-KF model performs better with respect to localization accuracy than the most advanced techniques.

Linda K. Ashim, Nimisha Suresh, et al. [3] compares several transfer learning models for the purpose of detecting skin cancer. Using previously trained models on new tasks with little data is known as transfer learning. The study assesses the performance of five models: Densenet, Xception, VGG16, EfficientNet, and Resnet—using a Kaggle dataset of skin cancer images. According to the research, VGG16 came in second with 81.94 accuracy, while Densenet earned the greatest accuracy at 88.61. The difficulties and potential paths for transfer learning in skin cancer diagnosis are also covered in the research. We were able to comprehend efficientnet's effectiveness by using the lessons we learned from this study project.

Shaik Mohammad Asif, T Anil, et al. [4] presents a deep learning-based method for traffic sign identification utilizing SSD and RCNN models. The study compares these models' performance to that of MobileNet, a lightweight CNN architecture that relies on depthwise separable convolutions. The models were evaluated using a Kaggle dataset of traffic signs, and the results showed that MobileNet achieved the optimum trade-off between accuracy and speed. The variables that affect detection accuracy are examined as well in the study, including image quality, rotation, occlusion, and weather. The study concludes that the MobileNet model is suitable for traffic sign recognition and detection in real-time.

Following a comprehensive review of the existing literature, we offer a comparative analysis of models that are precisely compatible with UAVs, or unmanned aerial vehicles. This connection facilitates real-time image processing and increases the effectiveness and efficiency of UAV operations.

3. Dataset

There is only one publicly available real world object detection dataset of side scan sonar images, in which each image consists of a single object, such as a drowning ship, airplane and pure seafloor, captured with the help of many sonar equipment suppliers, including Lcocean, Hydro-tech Marine, Klein Marine, Tritech, and EdgeTech. There are presently 487 ship images, 66 plains images and 583 seabed images in the collection known as SeabedObjects-KLSG-II. [5]. A few examples from the SeabedObjects-KLSG-II dataset is shown in Fig. 1. It is worth noting that each type of objects has its appearance diversity in Fig. 1, e.g., wrecks of different size and appearance seafloor of different types (the seafloor images from left to right are rock, mud, sand, sand waves, sand ridges, and clay, respectively).

4. Proposed Methodology

4.1. Data Preparation

Before the side-scan sonar (SSS) images were utilized for training both EfficientNet and MobileNet models, they had to undergo a variety of pre-processing steps. Among these were labeling, center cropping, resizing, and normalization. When center cropping, the image's center served as the reference point. As a result, the bottom had less of an influence, and the focus was more on the thing that was submerged. The sonar images were clipped and then uniformly resized

to 224 pixels wide by 224 pixels high in order to satisfy the input size criteria of the EfficientNet / MobileNet models. All of the dataset's images, which came with 8-bit, 24-bit, and 32-bit formats, were converted to normal 24-bit RGB images in order to meet the input size requirements. The labels for the images inside the folders were the folder names. In order to train the EfficientNet and MobileNet models, this got the SSS data ready. The data from each of the 3 channels within the SSS image was then normalized and placed in the range [-1, 1]. This crucial step was necessary to prevent results from training that were suboptimal due to significant data disparities. .

4.2. Data Augmentation

To improve the performance of the EfficientNet and MobileNet models, data augmentation strategies were employed during the training phase. Each time the model was trained, the input image underwent random transformations using methods that included random horizontal flip, random rotation (-50°, 50°), and random translations in both horizontal and vertical directions. Images of planes were assigned more weight to increase their presence and representation in the dataset. This approach not only enhanced the models' accuracy but also broadened the variety of the original data.

4.3. Splitting Training and Testing Data-sets

The training and testing databases of ship and aircraft sonar images were separated. Thirty percent of the images in each category were utilized for testing, while seventy percent were used for training. The EfficientNet and MobileNet models were trained on the training dataset, and their accuracy was assessed on the testing dataset.

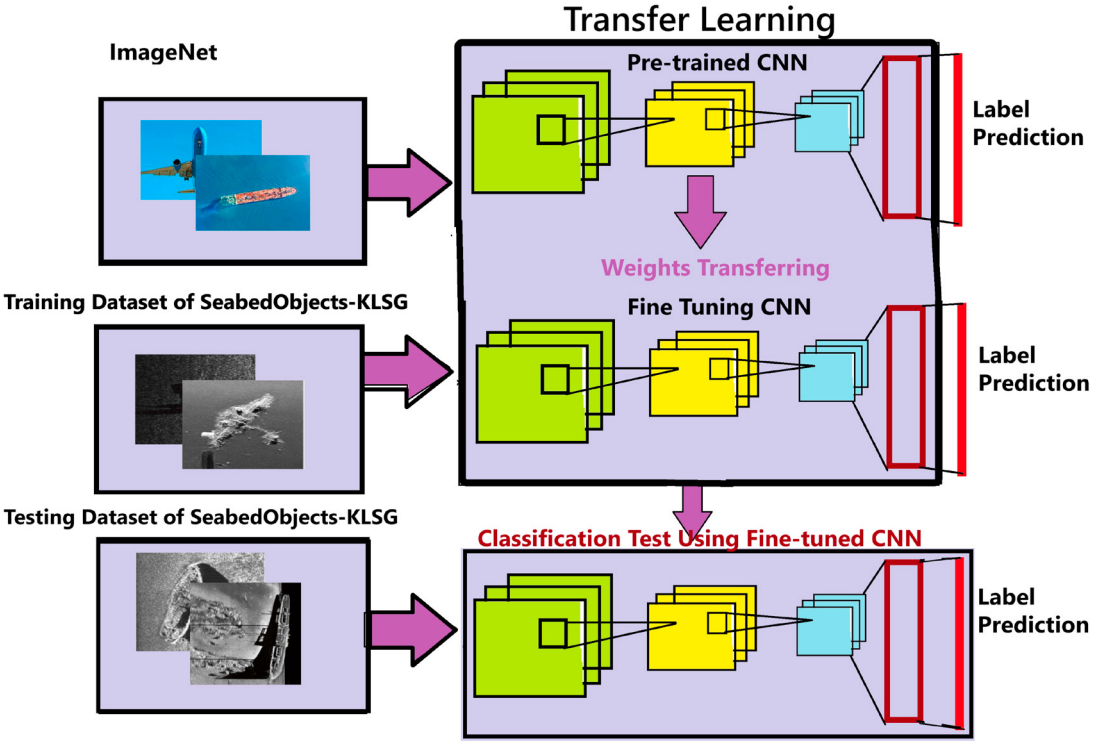


Fig. 2: Proposed System Architecture

4.4. Model Evaluation

We evaluated two key aspects in this work: the impact of transfer learning on model accuracy and the relative accuracy of the EfficientNet and MobileNet models. We also assessed the computational efficiency of the models. To determine

how transfer learning affects the accuracy of target recognition in side-scan sonar (SSS) images, we conducted training and prediction tests for both CNN models—EfficientNet and MobileNet—with and without transfer learning. Using the standard evaluation parameter, accuracy, we compared the accuracy of the two models chosen for our study. For the purposes of our study, ships and airplanes in underwater images were considered fair targets.

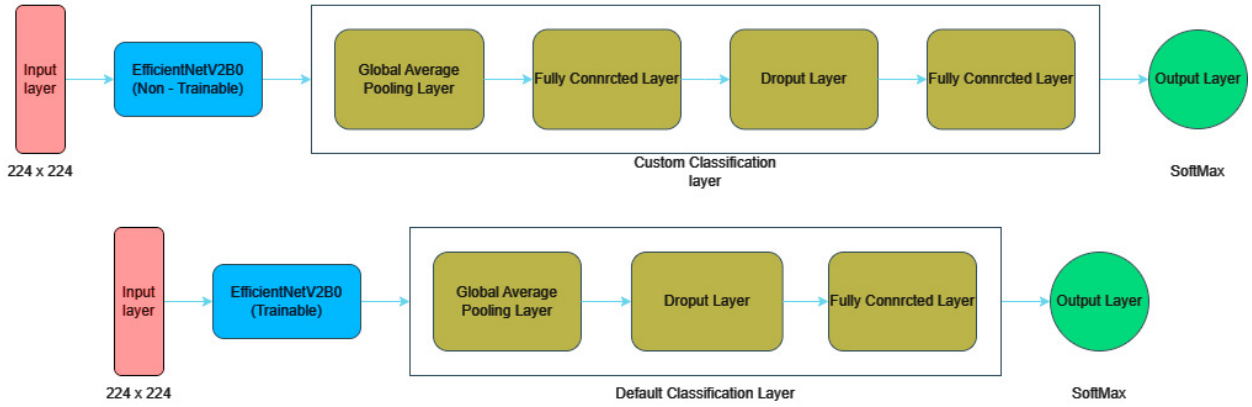


Fig. 3: Architecture of EfficientNet

4.5. Applied CNN Models

In our research on underwater object classification using side-scan sonar images, we are exploring the practicality and advantages of employing two distinct Convolutional Neural Network (CNN) models: EfficientNet [8][13] and MobileNet [10][15.EfficientNet's scalability allows precise tailoring of the model size, balancing computational demands and classification performance in resource-limited underwater scenarios. This feature is particularly valuable for efficient CNN usage in challenging environments.MobileNet's low-weight architecture makes it suitable for real-time applications, crucial for autonomous underwater vehicles. Its computational efficiency excels in scenarios where quick decision-making with minimal processing overhead is essential.This study aims to determine the efficacy of MobileNet and EfficientNet, assessing their advantages and disadvantages in underwater object classification. Results will contribute insights into their relative merits and help identify which model, if any, addresses challenges in categorizing underwater objects from side-scan sonar data. The study also compares transfer learning against using the basic model, providing valuable insights into its pros and cons as shown in Fig. 2.

4.6. CNN's Establishment

In this study, both EfficientNet and MobileNet underwent training with random weights added to their default top layers. As shown in Fig. 3 and Fig. 4 the output layer utilized a Softmax activation function for identifying three classes and assessing model accuracy. Transfer learning was then applied, utilizing pre-trained parameters from the ImageNet database. The top layer was replaced with a traditional output layer, featuring Softmax activation for classification and two fully connected dense layers with ReLU activation. This custom layer served as the classifier, while the baseline model acted as the feature extractor. Multiple parameter adjustments were made to optimize both models, resulting in four trained CNN models. The main goals were to evaluate model quality and understand the impact of pre-training variables on accuracy.

5. Experimental Results

Thoroughly examining the primary findings in this section is one facet of our study project. The primary discoveries include the effect of transfer learning on model precision, the precision of distinct CNN models in forecasting SSS

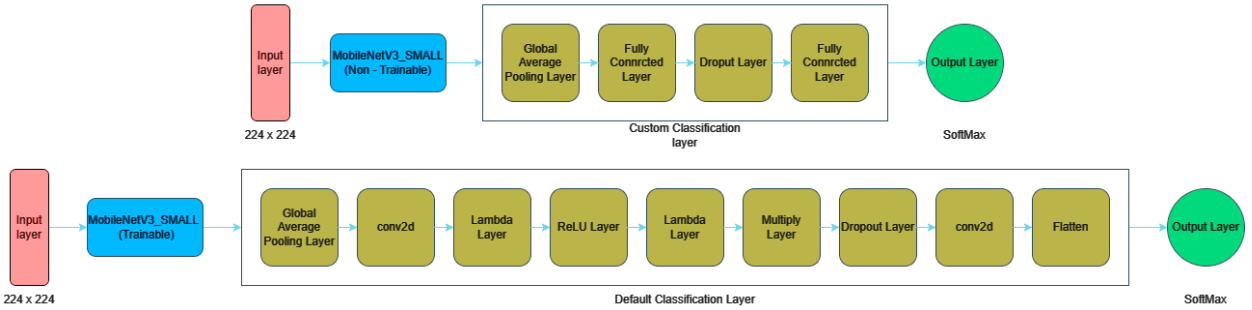


Fig. 4: Architecture of MobileNet

(SideScan Sonar) images, and the computational effectiveness of different models.To evaluate the accuracy of different CNN models in recognizing targets within SSS images, we conducted a comparative study both before and after applying transfer learning, as shown in Fig. 5. It illustrates the accuracy variation with respect to training epochs for target prediction in SSS images using the four CNN models, both before and after the application of transfer learning. Remarkably, prediction accuracy experiences a significant improvement after implementing transfer learning. These results underscore the substantial enhancement in training accuracy that transfer learning brings to CNN models. However, the degree of improvement varies from one model to another as depicted in Table. 1. To better understand the effects of transfer learning on different models, we have gathered information on the training and prediction performances. Notably, both EfficientNet and MobileNet have incorporated transfer learning, resulting in a significant

Table 1: Comparison before and after Using Transfer Learning

Feature	**MobileNet V3**	**EfficientNet B2**
Training Accuracy	99.35	98.70
Testing Accuracy	90.98	90.12
Training Accuracy when transfer learning is used	100	99.92
Testing Accuracy when transfer learning is used	97.25	99.05

improvement in accuracy on the testing dataset. However, the accuracy gain on the training dataset is somewhat small. This finding suggests that the conventional CNN structure's design is less affected by transfer learning and has a lesser impact on the training dataset's correctness. On the other hand, both models demonstrated significant accuracy gains on the test dataset with their innovative network topologies after implementing transfer learning.For example, EfficientNet exhibited a modest increase in the training dataset's accuracy, elevating it by 1.22 from 90.12 to 99.05. Although the test dataset's accuracy also increased, transfer learning was not essential to sustain the improvement. This could be attributed to EfficientNet's initial training performance of 98.70, which was already quite robust. MobileNet, on the other hand, showed an improvement of around 6.27 on the test dataset. A comparison of EfficientNet with MobileNet's performance revealed some interesting results. MobileNet demonstrated quicker convergence to the best gradient, providing an advantage in terms of training-related aspects. However, when it came to accuracy, EfficientNet outperformed MobileNet, boasting a greater accuracy rate of 2.20. The lightweight design and adjustable features of MobileNet contribute to its faster convergence and fewer trainable parameters, potentially explaining its advantage. Therefore, in scenarios with less data, MobileNet performs better, while EfficientNet is ultimately more accurate and adaptable to various datasets, prioritizing economy and overall performance.

5.1. Loss Graph

Analyzing the loss graphs revealed further numerical information.Fig 7 shows in just 20 epochs, MobileNetV3 demonstrated a fast convergence and reached a training loss of 0.1, while in Fig 6 EfficientNetV2B0 reached a similar loss by the 25th epoch. Nevertheless, as the validation loss diverged and reached 0.3 at the 30th epoch, MobileNetV3 showed

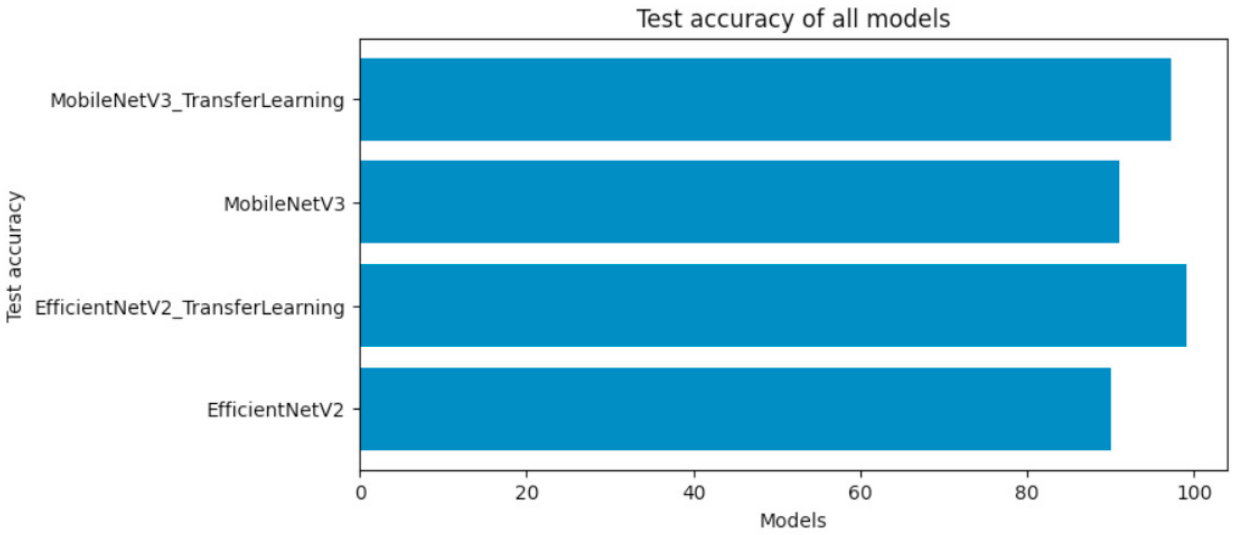


Fig. 5: Accuracy comparison

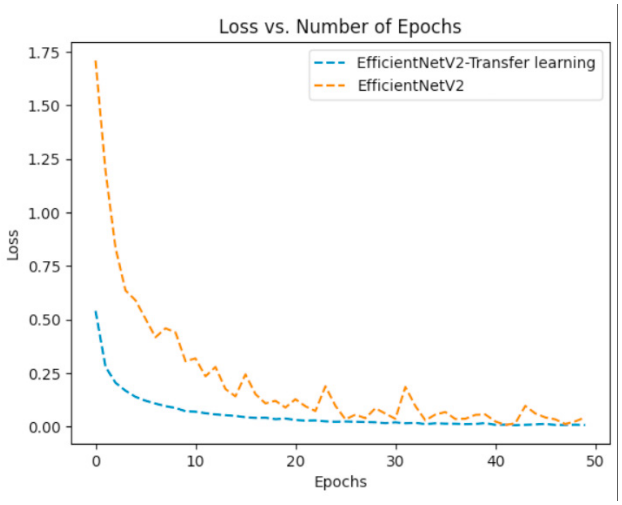


Fig. 6: EfficientNet Loss Graph

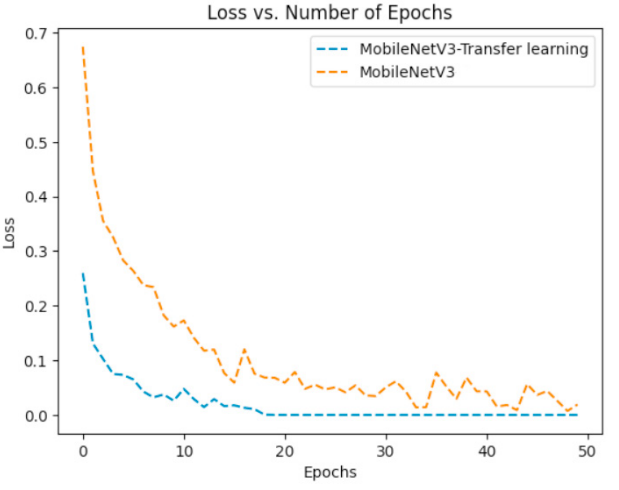


Fig. 7: MobileNet Loss Graph

indications of possible overfitting. EfficientNetV2B0, on the other hand, continued to operate steadily, with the validation loss roughly mirroring the training loss. These numerical findings illustrate the subtle trade-offs between the models: EfficientNetV2B0 showed a stable and steadily improving performance, while MobileNetV3 demonstrated faster convergence but was susceptible to overfitting.

5.2. Accuracy Graph

Analyzing the accuracy graphs of MobileNetV3 as shown in Fig. 9 and EfficientNetV2 as shown in Fig. 8 revealed interesting patterns. Initially, MobileNetV3 led with an accuracy of 85%, but its development reached a standstill after 25 epochs. In contrast, EfficientNetV2 B0 demonstrated a more steady and consistent rise, surpassing MobileNetV3 with an accuracy of 88% at the 30th epoch. This numerical representation illustrates how EfficientNetV2 B0 outperformed MobileNetV3 by maintaining a consistent and increasing accuracy improvement over time.

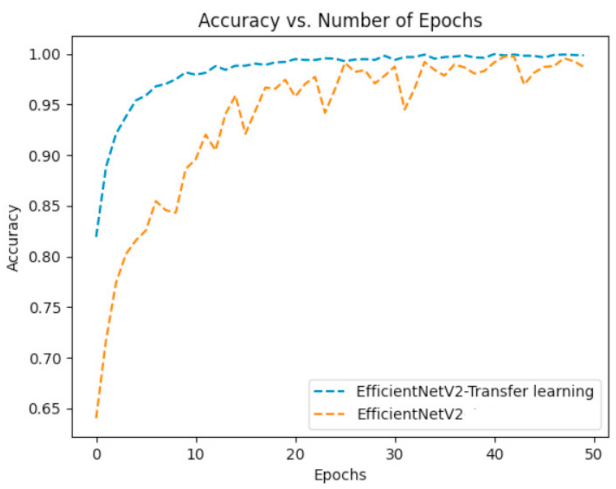


Fig. 8: EfficientNet Accuracy Graph

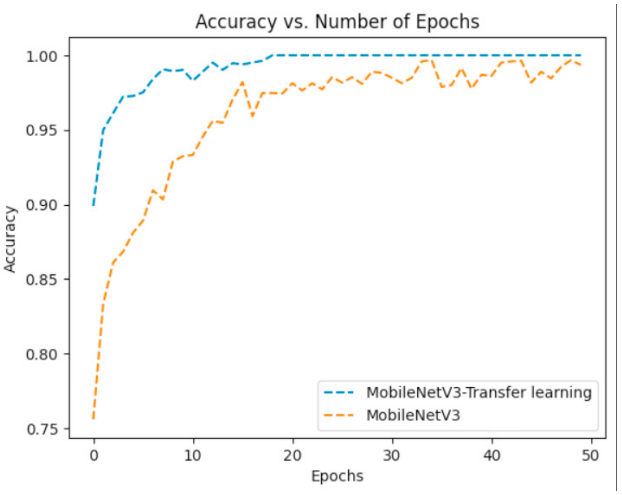


Fig. 9: MobileNet Accuracy Graph

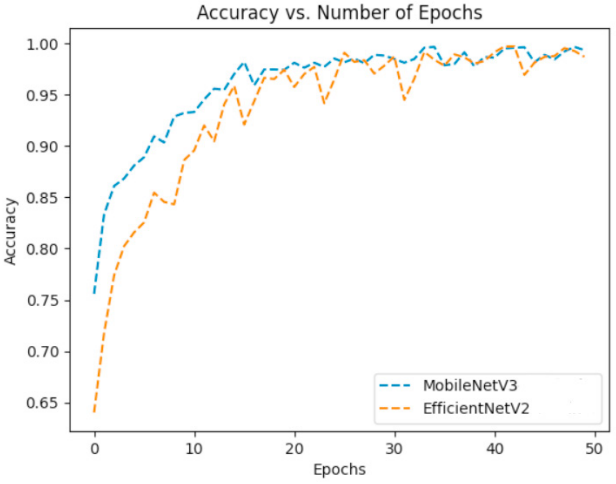


Fig. 10: Accuracy Graph

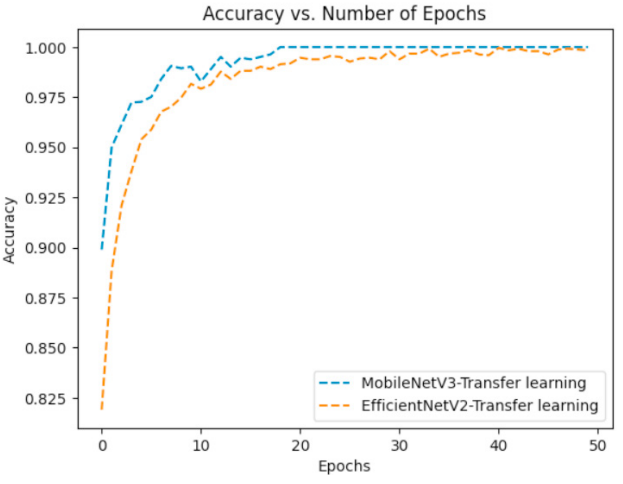


Fig. 11: Transfer Learning Accuracy Graph

5.3. Model Comparison

Transfer Accuracy Using EfficientNet is the most effective approach to efficiency when compared to other approaches as seen in Fig. 10 and Fig. 11. MobileNet is a great option if you prefer a very light and efficient variant. This results can be reinforced be comparing the loss observed between the 2 models as shown in Fig. 12 and Fig. 13. The conditions given in Table. 2 might influence the choice you make when selecting between these two models:

This study briefly compares the computational effectiveness and quantity of network parameters of the CNN models used. The intricacy of the CNN network structure is reflected in its layers and the number of parameters.

Additional layers on the CNN network enable it to handle more complex image processing issues. As the number of parameters and Multiply-Accumulate operations (MACs) increases, the required hardware for computation grows and becomes more sophisticated. The main measures used to evaluate computational efficiency are the total number of network layers and calculation time.The base model MobileNet starts with just 88 layers, while MobileNetV3_Small at alpha=1.0, used in this study, contains 228 layers with a reasonable number of parameters and MACs. It calculates quickly as well. EfficientNet has a variable number of layers depending on the scaling ratio, frequently containing

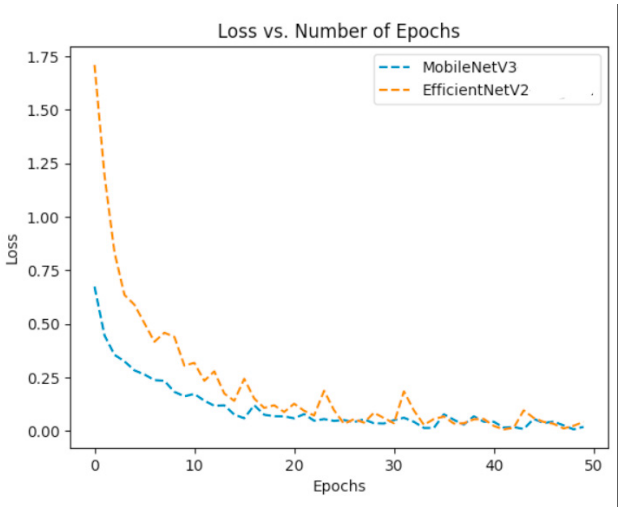


Fig. 12: Loss Graph

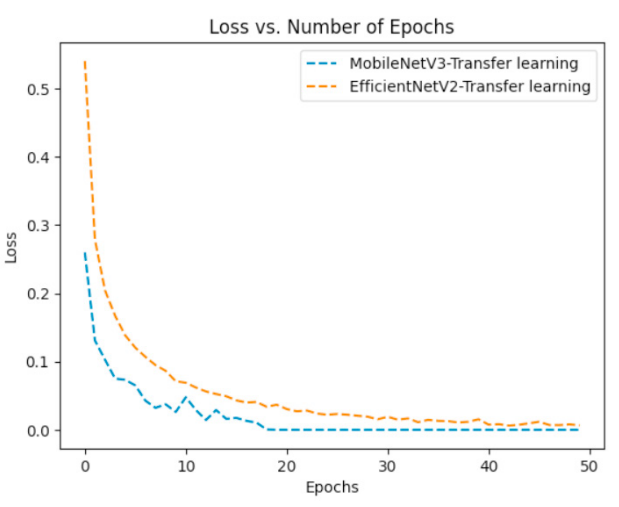


Fig. 13: Transfer Learning Loss Graph

Table 2: Model Comparison

Feature	**MobileNetV3small**	**EfficientNetV2B0**
Model Type	Light Weight	Medium weight
Number of parameters	2.9M	7.2M
MACs	66	29
Top-1 accuracy on ImageNet	68.1	78.7
Training accuracy increases when transfer learning is applied.	0.65	1.22
Test accuracy increases when transfer learning is applied.	6.27	8.93

more parameters and layers. Both models use a novel CNN structure that employs depth-wise differentiated convolutions and compound scaling to achieve high accuracy and efficiency for various image applications. EfficientNet, known for scalability, achieved an impressive accuracy of approximately 99.05%, demonstrating robust performance even in challenging underwater conditions. On the other hand, MobileNet, with its lightweight design, exhibited a consistent accuracy of around 97%, making it particularly suitable for applications requiring fast decision-making, such as autonomous underwater vehicles. The comparison highlights the strengths and trade-offs of these models, providing valuable insights for the effective deployment of sonar image detection in underwater environments.

6. Conclusion

In conclusion, this paper addresses the challenges in deep learning, particularly when dealing with limited resources. The performance of various computer models was compared, with a focus on the trade-off between the number of layers and computational speed. MobileNet and MobileNetV3_Small, with 88 and 228 layers, respectively, showcased a balance between details and speed. In contrast, EfficientNet, with its adjustable number of layers, generally offers more detailed information, making it effective for handling complex images. However, both EfficientNet and MobileNet perform optimally with specialized computer assistance.Future research should concentrate on enhancing models for real-time deployment while considering resource constraints, especially in submerged vehicles or remote sensing devices. Optimization efforts could involve techniques such as model quantization, pruning, and other measures to improve the efficiency of the inference process. Exploring multimodal fusion to integrate additional data from sensors, such as pressure, temperature, or electromagnetic readings, with sonar imaging could further enhance the comprehensive understanding of the underwater environment. Additionally, integrating anomaly detection into the project could train models to recognize not only conventional underwater structures but also unusual or unexpected

patterns. This enhancement would enable the models to identify potential threats or intriguing events in the underwater world.

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References

[1] Du, Xing, Yongfu Sun, Yupeng Song, Huifeng Sun, and Lei Yang. 2023. "A Comparative Study of Different CNN Models and Transfer Learning Effect for Underwater Object Classification in Side-Scan Sonar Images" Remote Sensing 15, no. 3: 593. https://doi.org/10.3390/rs15030593

[2] Sreekala, Keshetti, N. Nijil Raj, Sachi Gupta, G. Anitha, Ashok Kumar Nanda, and Abhay Chaturvedi. "Deep Convolutional Neural Network with Kalman Filter Based Objected Tracking and Detection in Underwater Communications." Wireless Networks, 2023. https://doi.org/10.1007/s11276-023-03290-z.

[3] Ashim, Linda K, Nimisha Suresh, and Prasannakumar C V. "A Comparative Analysis of Various Transfer Learning Approaches Skin Cancer Detection." 2021 5th International Conference on Trends in Electronics and Informatics (ICOEI), 2021. https://doi.org/10.1109/icoei51242.2021.9452854.

[4] Asif, Shaik Mohammad, T Anil, Satwik Tangudu, GH Sai Keertan, and Don S. "Traffic Sign Detection Using SSD Mobilenet & Faster RCNN." 2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), 2023. https://doi.org/10.1109/vitecon58111.2023.10157917.

[5] Huo, Guanying, Ziyin Wu, and Jiabiao Li. "Underwater Object Classification in Sidescan Sonar Images Using Deep Transfer Learning and Semisynthetic Training Data." IEEE Access 8 (2020): 47407–18. https://doi.org/10.1109/access.2020.2978880.

[6] Phung, S. L., T. N. Nguyen, H. T. Le, P. B. Chapple, C. H. Ritz, A. Bouzerdoum, and L. C. Tran. "Mine-like Object Sensing in Sonar Imagery with a Compact Deep Learning Architecture for Scarce Data." 2019 Digital Image Computing: Techniques and Applications (DICTA), 2019. https://doi.org/10.1109/dicta47822.2019.8945982.

[7] Rao, Chinmay, Kushal Mukherjee, Shalabh Gupta, Asok Ray, and Shashi Phoha. "Underwater Mine Detection Using Symbolic Pattern Analysis of Sidescan Sonar Images." 2009 American Control Conference, 2009. https://doi.org/10.1109/acc.2009.5160102.

[8] Yadav, Pooja, Neeraj Menon, Vinayakumar Ravi, Sowmya Vishvanathan, and Tuan D. Pham. "EfficientNet Convolutional Neural Networks-Based Android Malware Detection." Computers & Security 115 (2022): 102622. https://doi.org/10.1016/j.cose.2022.102622.

[9] Feng, Sheng, and Xiaoqian Zhu. "A Transformer-Based Deep Learning Network for Underwater Acoustic Target Recognition." IEEE Geoscience and Remote Sensing Letters 19 (2022): 1–5. https://doi.org/10.1109/lgrs.2022.3201396.

[10] C M, Sruthy, Sandra Kunjumon, and Nandakumar R. "Car Damage Identification and Categorization Using Various Transfer Learning Models." 2021 5th International Conference on Trends in Electronics and Informatics (ICOEI), 2021. https://doi.org/10.1109/icoei51242.2021.9452846.

[11] Thanh Le, Hoang, Son Lam Phung, Philip B. Chapple, Abdesselam Bouzerdoum, Christian H. Ritz, and Le Chung Tran. "Deep Gabor Neural Network for Automatic Detection of Mine-like Objects in Sonar Imagery." IEEE Access 8 (2020): 94126–39. https://doi.org/10.1109/access.2020.2995390.

[12] M, Bindhumol, Tripty Singh, and Prakash Duraisamy. "Diabetic Retinopathy Classification Using Transfer Learning Techniques." 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT), 2022. https://doi.org/10.1109/icccnt54827.2022.9984622.

[13] Xi, Caiping, and Renqiao Liu. 2022. "Detection of Small Floating Target on Sea Surface Based on Gramian Angular Field and Improved EfficientNet" Remote Sensing 14, no. 17: 4364. https://doi.org/10.3390/rs14174364

[14] Huang, Xiaochao, Rong Yang, Qi Wang, Fei Yu, and Bo He. "A Novel Method for Real-Time ATR System of AUV Based on Attention-Mobilenetv3 Network and Pixel Correction Algorithm." Ocean Engineering 270 (2023): 113403. https://doi.org/10.1016/j.oceaneng.2022.113403.

[15] Zhang, Yulong, Jingtao Sun, Mingkang Chen, and Qiang Wang. "Efficient Semantic Segmentation Backbone Evaluation for Unmanned Surface Vehicles Based on Likelihood Distribution Estimation." 2022 18th International Conference on Mobility, Sensing and Networking (MSN), 2022. https://doi.org/10.1109/msn57253.2022.00076.